

PNETLab | Topology

Notes/CCNA/SN-CN-B1/Static

Invalid Next Hop Error

Ask Gemini

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Notifications

QEMU (Win-1)

C:\WINDOWS\system32\cmd.exe

Ping statistics for 30.0.0.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 1ms, Average = 1ms
C:\Documents and Settings\User>ping 40.0.0.10
Pinging 40.0.0.10 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Ping statistics for 40.0.0.10:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\Documents and Settings\User>ping 40.0.0.10
Pinging 40.0.0.10 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Ping statistics for 40.0.0.10:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\Documents and Settings\User>ping 40.0.0.10
Pinging 40.0.0.10 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Ping statistics for 40.0.0.10:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\Documents and Settings\User>ping 40.0.0.10
Pinging 40.0.0.10 with 32 bytes of data:
Reply from 40.0.0.10: bytes=32 time=3ms TTL=125
Reply from 40.0.0.10: bytes=32 time=1ms TTL=125
Reply from 40.0.0.10: bytes=32 time=2ms TTL=125
Reply from 40.0.0.10: bytes=32 time=1ms TTL=125
Ping statistics for 40.0.0.10:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 3ms, Average = 1ms
C:\Documents and Settings\User>

Win-1
10.0.0.10

Win-2
40.0.0.10

Static Route

StartNetwork ConnectionsC:\WINDOWS\system32...8:46 PM01 : 00 : 00START

Search1:16 AM7/16/2026

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Notifications

QEMU (Win-2)

```

C:\WINDOWS\system32\cmd.exe
Pinging 40.0.0.1 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Ping statistics for 40.0.0.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\Documents and Settings\user>ping 10.0.0.10
Pinging 10.0.0.10 with 32 bytes of data:
Reply from 10.0.0.10: bytes=32 time=4ms TTL=125
Reply from 10.0.0.10: bytes=32 time=2ms TTL=125
Reply from 10.0.0.10: bytes=32 time=2ms TTL=125
Reply from 10.0.0.10: bytes=32 time=1ms TTL=125
Ping statistics for 10.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 4ms, Average = 2ms
C:\Documents and Settings\user>

```

Win-1 10.0.0.10

Cisco ASDM-1... Old Java ASDM

PUTTY

Trpd32

Win-2 40.0.0.10

Static Route

Start Network Connections C:\WINDOWS\system... 8:47 PM 01 : 00 : 00 START

Search

1:17 AM 7/16/2026

```
PutTY (inactive)
R2#
R2#
R2#
R2#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S    10.0.0.0/8 [1/0] via 20.0.0.1
    20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    20.0.0.0/8 is directly connected, Ethernet0/0
L    20.0.0.2/32 is directly connected, Ethernet0/0
C    30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    30.0.0.0/8 is directly connected, Ethernet0/1
L    30.0.0.1/32 is directly connected, Ethernet0/1
S    40.0.0.0/8 [1/0] via 30.0.0.2
R2#
```

```
Putty (inactive)
R1(config)#
R1(config)#exit
R1#
*Jul 15 19:32:37.664: %SYS-5-CONFIG_I: Configured from console by console
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
        a - application route
        + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

  10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/8 is directly connected, Ethernet0/0
L       10.0.0.1/32 is directly connected, Ethernet0/0
C       20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       20.0.0.0/8 is directly connected, Ethernet0/1
L       20.0.0.1/32 is directly connected, Ethernet0/1
S       30.0.0.0/8 [1/0] via 20.0.0.2
S       40.0.0.0/8 [1/0] via 20.0.0.2
R1#
```

```
R3
Ethernet0/2          unassigned      YES unset  administratively down down
Ethernet0/3          unassigned      YES unset  administratively down down

Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int e0/0
Router(config-if)#ip add 40.0.0.1 255.0.0.0
Router(config-if)#no sh
Router(config-if)#e1
*Jul 15 19:44:59.327: %LINK-3-UPDOWN: Interface Ethernet0/0, changed state to up
Router(config-if)#e1t
*Jul 15 19:45:00.326: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/0, changed state to up
Router(config-if)#ex
Router(config)#ex
% Ambiguous command: "ex"
Router(config)#exit
Router#
*Jul 15 19:45:13.948: %SYS-5-CONFIG_I: Configured from console by console
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip route 10.0.0.0 255.0.0.0 30.0.0.1
Router(config)#ip route 20.0.0.0 255.0.0.0 30.0.0.1
Router(config)#exit
Router#
*Jul 15 19:46:11.575: %SYS-5-CONFIG_I: Configured from console by console
Router#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S    10.0.0.0/8 [1/0] via 30.0.0.1
S    20.0.0.0/8 [1/0] via 30.0.0.1
    30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    30.0.0.0/8 is directly connected, Ethernet0/1
L    30.0.0.2/32 is directly connected, Ethernet0/1
    40.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    40.0.0.0/8 is directly connected, Ethernet0/0
L    40.0.0.1/32 is directly connected, Ethernet0/0
Router#
```

```
R2
*Jul 15 19:05:23.556: %IP-4-DUPADDR: Duplicate address 30.0.0.1 on Ethernet0/1,
sourced by aabb.cc00.0310
R2#
*Jul 15 19:05:53.556: %IP-4-DUPADDR: Duplicate address 30.0.0.1 on Ethernet0/1,
sourced by aabb.cc00.0310
R2#
*Jul 15 19:06:23.556: %IP-4-DUPADDR: Duplicate address 30.0.0.1 on Ethernet0/1,
sourced by aabb.cc00.0310
R2#
*Jul 15 19:06:53.556: %IP-4-DUPADDR: Duplicate address 30.0.0.1 on Ethernet0/1,
sourced by aabb.cc00.0310
R2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#
*Jul 15 19:07:23.556: %IP-4-DUPADDR: Duplicate address 30.0.0.1 on Ethernet0/1,
sourced by aabb.cc00.0310
R2(config)#ip route 10.0.0.0 255.0.0.0 20.0.0.1
*Jul 15 19:07:53.556: %IP-4-DUPADDR: Duplicate address 30.0.0.1 on Ethernet0/1,
sourced by aabb.cc00.0310
R2(config)#ip route 10.0.0.0 255.0.0.0 20.0.0.1
R2(config)#
R2#
*Jul 15 19:08:15.894: %SYS-5-CONFIG_I: Configured from console by console
R2#sh ip route
*Jul 15 19:08:23.556: %IP-4-DUPADDR: Duplicate address 30.0.0.1 on Ethernet0/1,
sourced by aabb.cc00.0310
R2#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
        a - application route
        + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S    10.0.0.0/8 [1/0] via 20.0.0.1
    20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    20.0.0.0/8 is directly connected, Ethernet0/0
L    20.0.0.2/32 is directly connected, Ethernet0/0
    30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    30.0.0.0/8 is directly connected, Ethernet0/1
L    30.0.0.1/32 is directly connected, Ethernet0/1
R2#
*Jul 15 19:08:53.556: %IP-4-DUPADDR: Duplicate address 30.0.0.1 on Ethernet0/1, sourced by aabb.cc00.0310
R2#
```